

# Preface

in terms of vectors

用A为插  
进线性组合

(1) two column vectors:  $\text{lin } \mathbb{R}^3$   
 $a_1 = \begin{bmatrix} 2 \\ 3 \\ 1 \end{bmatrix}$   $a_2 = \begin{bmatrix} 1 \\ 4 \\ 2 \end{bmatrix}$

(2) linear combination:  $ca_1 + da_2 \iff$  fill a whole plane

↓ 将  $a_1, a_2$  集成! ↑

(3) Matrix  $A$  holds  $n$  column vectors  $a_1 \dots a_n$  等价  
 e.g.  $A = [a_1 a_2] = \begin{bmatrix} 2 & 1 \\ 3 & 4 \\ 1 & 2 \end{bmatrix}$

(in term of matrix  $A$ !)

$ca_1 + da_2$ : the column space of the matrix.

(4) include 2 more columns

$$A = \begin{bmatrix} 2 & 1 & 3 & 0 \\ 3 & 4 & 7 & 0 \\ 1 & 2 & 3 & 1 \end{bmatrix}$$

$\Rightarrow A$ 's column space: 3D  
 raises or lowers the plane

not on the plane.

$a_1 + a_2 \Rightarrow$  contribute to nothing new

$\Rightarrow$  columns:  $\begin{cases} \text{independent} \\ \text{dependent} \end{cases}$

(5) 5.1  $A = CR \Rightarrow$  拆解 dependent 属性.

$C$ : the independent columns (相对的).

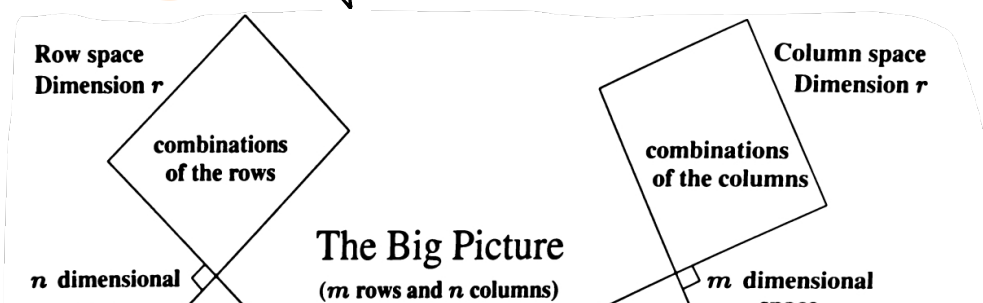
$R$ : 线性组合系数矩阵

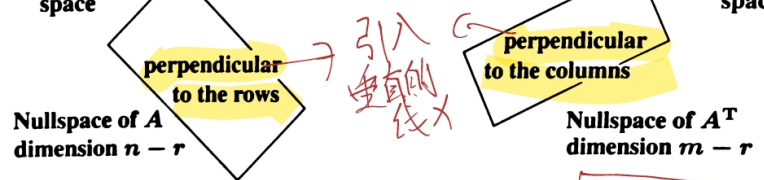
$$\text{e.g. } A = \begin{bmatrix} 2 & 1 & 3 & 0 \\ 3 & 4 & 7 & 0 \\ 1 & 2 & 3 & 1 \end{bmatrix} = \begin{bmatrix} 2 & 1 & 0 \\ 3 & 4 & 0 \\ 1 & 2 & -1 \end{bmatrix} \cdot \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \end{bmatrix} = CR.$$

5.2: matrix multiplication: column  $j$  of  $CR$  is  $C$  times column  $j$  of  $R$ !

(6) matrix analysis:

{ column space  
row space.





The Four Fundamental Subspaces: An  $m$  by  $n$  matrix with  $r$  independent columns.

## (6) Matrix Factorization:

"When our matrix has a special property, these factorizations will show it"

6-1 Here are the five factorizations from Chapters 1, 2, 4, 6, 7:

- 1  $A = CR$  =  $R$  combines independent columns in  $C$  to give all columns
- 2  $A = LU$  = Lower triangular  $L$  times Upper triangular  $U$
- 4  $A = QR$  = Orthogonal matrix  $Q$  times Upper triangular  $R$
- 6  $S = Q\Lambda Q^T$  = (Orthogonal  $Q$ ) (Eigenvalues in  $\Lambda$ ) (Orthogonal  $Q^T$ )
- ★ 7  $A = U\Sigma V^T$  = (Orthogonal  $U$ ) (Singular values in  $\Sigma$ ) (Orthogonal  $V^T$ )

最简单的正交矩阵例子  $\leftarrow$  正交矩阵  $\Rightarrow$  算法基石.

e.g.  $\begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} = \text{rotation by angle } \theta$

$A \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} \cos\theta \\ \sin\theta \end{bmatrix}$  column vector  $\perp$  且长度为 1

e.g.  $A$  为: 旋转后特向量的坐标